

Lecture 7:

Large Language Models

CS6493 Natural Language Processing
Instructor: Linqi Song

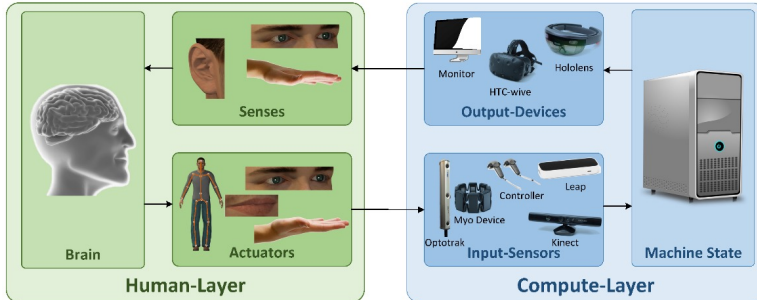


Outline

- Large language model definition
- Scaling law and emergent abilities
- Alignment

ChatGPT becomes the the fastest-growing 'app' of all time

- ChatGPT, Chat Generative Pre-trained Transformer, is an artificial intelligence (AI) chatbot developed by OpenAI and released in November 2022.
- ChatGPT can communicate with human beings in a more **natural way**.



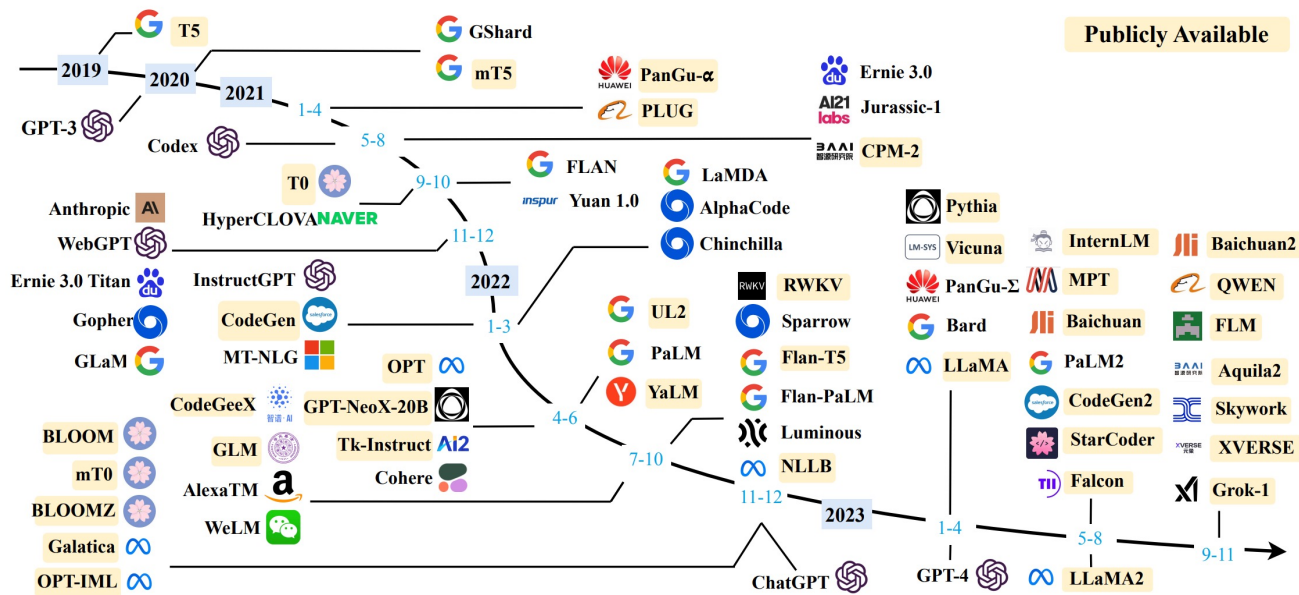
☀ Examples	⚡ Capabilities	⚠ Limitations
"Explain quantum computing in simple terms" →	Remembers what user said earlier in the conversation	May occasionally generate incorrect information
"Got any creative ideas for a 10 year old's birthday?" →	Allows user to provide follow-up corrections	May occasionally produce harmful instructions or biased content
"How do I make an HTTP request in Javascript?" →	Trained to decline inappropriate requests	Limited knowledge of world and events after 2021

Send a message... 

ChatGPT Mar 23 Version. Free Research Preview. ChatGPT may produce inaccurate information about people, places, or facts

LLMs have powerful textual processing abilities

- LLMs can write and debug computer programs, mimic the style of celebrity CEOs and write business pitches, compose music, teleplays, fairy tales and student essays, answer test questions (sometimes, depending on the test, at a level above the average human test-taker), write poetry and song lyrics, emulate a Linux system; simulate entire chat rooms, play games like tic-tac-toe and simulate an ATM.



Example of LLM (1)

- Chatbot

 ChatGPT 

tell me a short story

...

Once upon a time, in a cozy little cottage nestled in the heart of a lush forest, lived a kind-hearted elderly woman named Amelia. Despite her age, Amelia possessed a youthful spirit and a relentless love for nature.

Every day, Amelia would wander into the forest, her walking stick in hand and a gentle smile on her face. She would greet the trees, sing to the birds, and marvel at the beauty that surrounded her. The animals of the forest soon recognized her as a friend and would playfully follow her on her daily adventures.

One sunny morning, as Amelia strolled along a winding path, she noticed a wounded bird with a broken wing huddled beneath a tree. Without hesitation, she gently picked up the bird, cradling it in her hands. With a soft touch and a tender heart, Amelia bound the bird's wing using a scrap of cloth.

 please tell me a English joke



Here's a simple English joke:

Why did the chicken cross the road?

To get to the other side!

It's a classic joke that plays on the expected punchline of "to get to the other side" when people ask "why did the chicken cross the road?" The joke is funny because it sounds serious at first, but then the unexpected and simple answer is revealed.

Example of LLM (2)

- Write code and solve math problem

write the bubble sort code in python

Sure, here is a simple implementation of the Bubble Sort algorithm in Python:

```
python
def bubble_sort(arr):
    n = len(arr)

    for i in range(n):
        for j in range(0, n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
        return arr

# Test the function
arr = [64, 34, 25, 12, 22, 11, 90]
print("Original Array:")
print(arr)

bubble_sort(arr)

print("\nSorted Array:")
print(arr)
```

Solve the problem:

Problem: In a farm, there is a cage that houses chickens and rabbits only. The total number of heads in the cage is 50, and the total number of feet is 120. Can you figure out how many chickens and how many rabbits are in the cage?

Note: Remember, a chicken has 1 head and 2 feet, and a rabbit has 1 head and 4 feet.

Sure, let's solve this problem.

Let's say the number of chickens is C and the number of rabbits is R.

From the problem, we know two things:

1. $C + R = 50$ (Because the total number of heads is 50)
2. $2C + 4R = 120$ (Because the total number of feet is 120)

We can simplify the second equation to $C + 2R = 60$ by dividing the whole equation by 2.

Now we have:

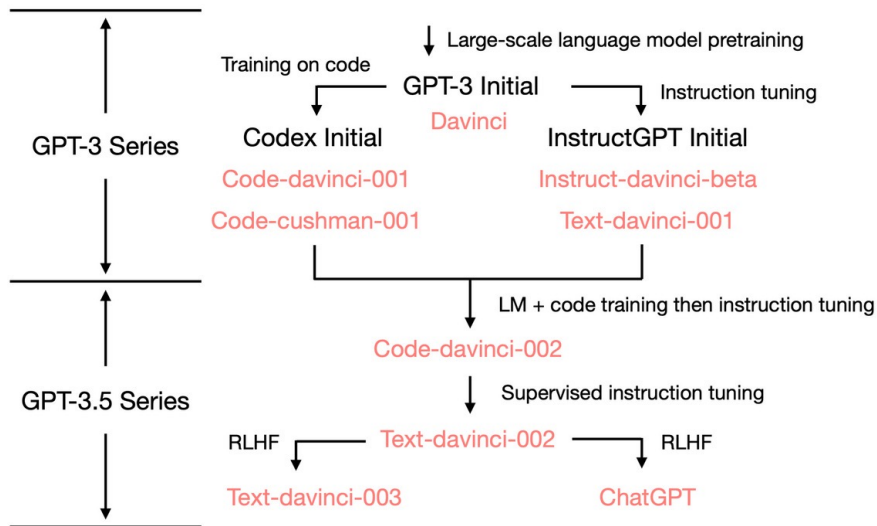
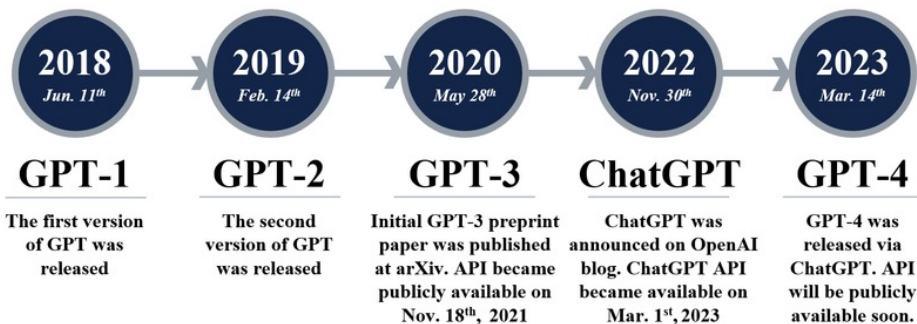
1. $C + R = 50$
2. $C + 2R = 60$

If we subtract the first equation from the second, we get $R = 10$.

Substituting $R = 10$ into the first equation, we get $C = 40$.

So, there are 40 chickens and 10 rabbits in the cage.

The evolution of GPT models



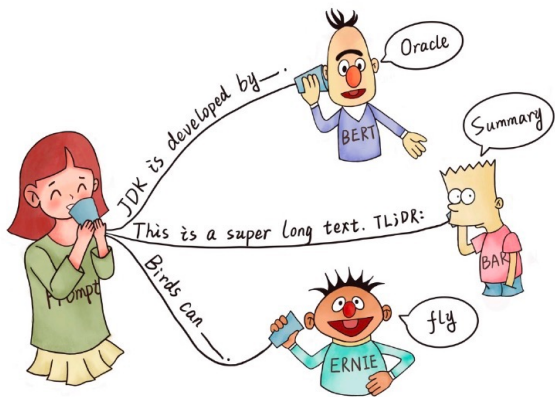
Comparisons between SLMs and LLMs

Aspect	Small Language Models	Large Language Models
Size	Can have less than 15 million parameters.	Can have hundreds of billions of parameters.
Computational Requirements	Can use mobile device processors.	Can require hundreds of GPU processors.
Performance	Can handle simple tasks.	Can handle complex, diverse tasks.
Deployment	Easier to deploy in resource-constrained environments.	Deployment often requires substantial infrastructure.
Training	Can be trained a week.	Training can take months.

Prompt engineering

A prompt is a **piece of text** inserted in the input examples, so that the original task can be formulated as a (masked) language modeling problem.

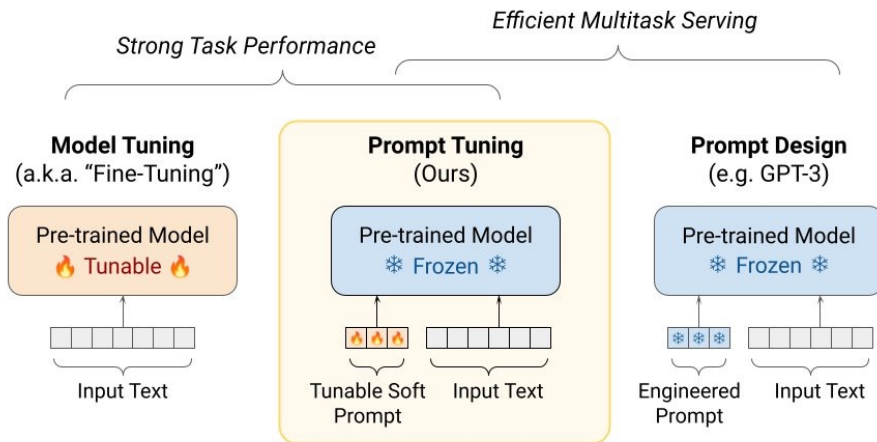
ChatGPT + Prompt



- Q&A**
Answer questions based on existing knowle...
- Summarize for a 2nd grader**
Translates difficult text into simpler concep...
- Text to command**
Translate text into programmatic commands.
- Natural language to Stripe API**
Create code to call the Stripe API using nat...
- Parse unstructured data**
Create tables from long form text
- Python to natural language**
Explain a piece of Python code in human un...
- Calculate Time Complexity**
Find the time complexity of a function.
- Advanced tweet classifier**
Advanced sentiment detection for a piece o...
- Keywords**
Extract keywords from a block of text.
- Ad from product description**
Turn a product description into ad copy.
- TL;DR summarization**
Summarize text by adding a 'tl;dr:' to the en...
- Spreadsheet creator**
Create spreadsheets of various kinds of dat...
- ML/AI language model tutor**
Bot that answers questions about language...
- Grammar correction**
Corrects sentences into standard English.
- Natural language to OpenAI API**
Create code to call to the OpenAI API usin...
- English to other languages**
Translates English text into French, Spanish...
- SQL translate**
Translate natural language to SQL queries.
- Classification**
Classify items into categories via example.
- Movie to Emoji**
Convert movie titles into emoji.
- Translate programming languages**
Translate from one programming language ...
- Explain code**
Explain a complicated piece of code.
- Factual answering**
Guide the model towards factual answering ...
- Product name generator**
Create product names from examples word...
- Python bug fixer**
Find and fix bugs in source code.
- JavaScript helper chatbot**
Message-style bot that answers JavaScript ...
- Science fiction book list maker**
Create a list of items for a given topic.

Paradigm shift

- **From pretraining-finetuning to prompting**
- **Pretraining-finetuning:** pretrained model is tunable and adapted downstream tasks
- **Prompting:** pretrained model is often frozen and some input prefix will be added to use the model to perform different tasks

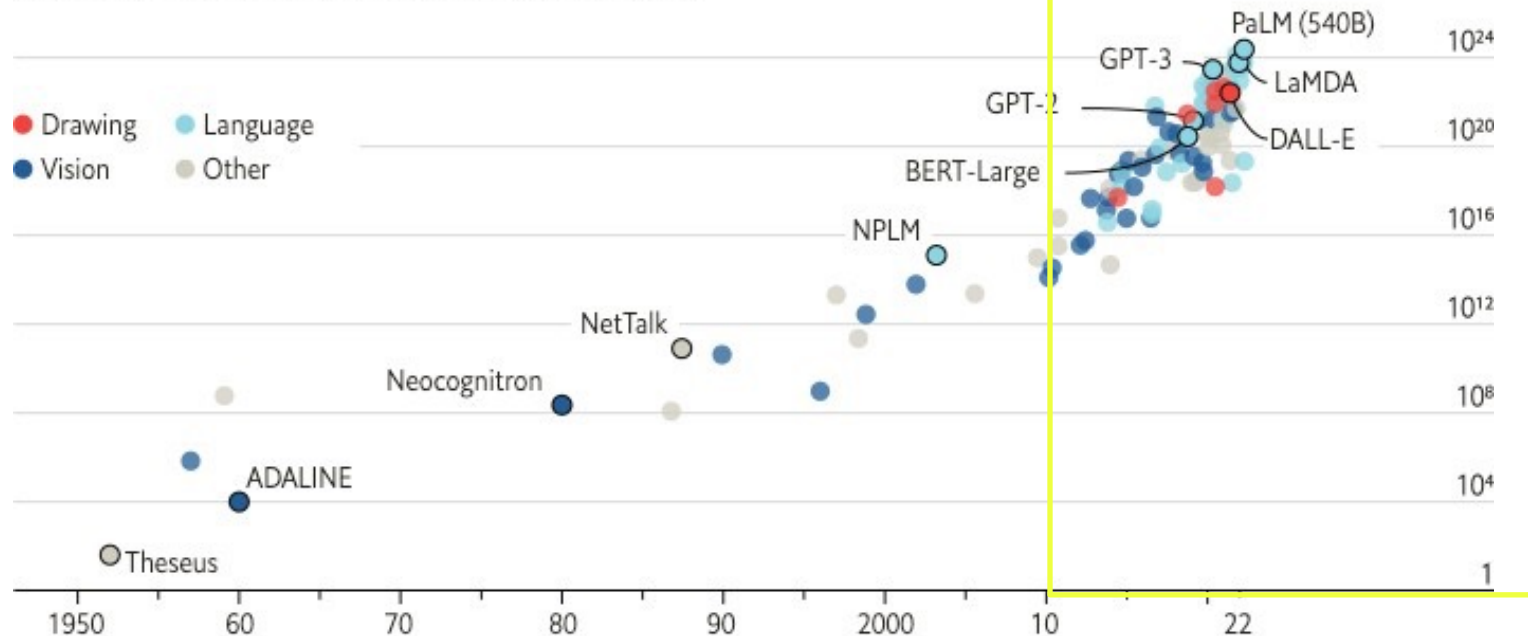


The era of large language models

The blessings of scale

AI training runs, estimated computing resources used

Floating-point operations, selected systems, by type, log scale

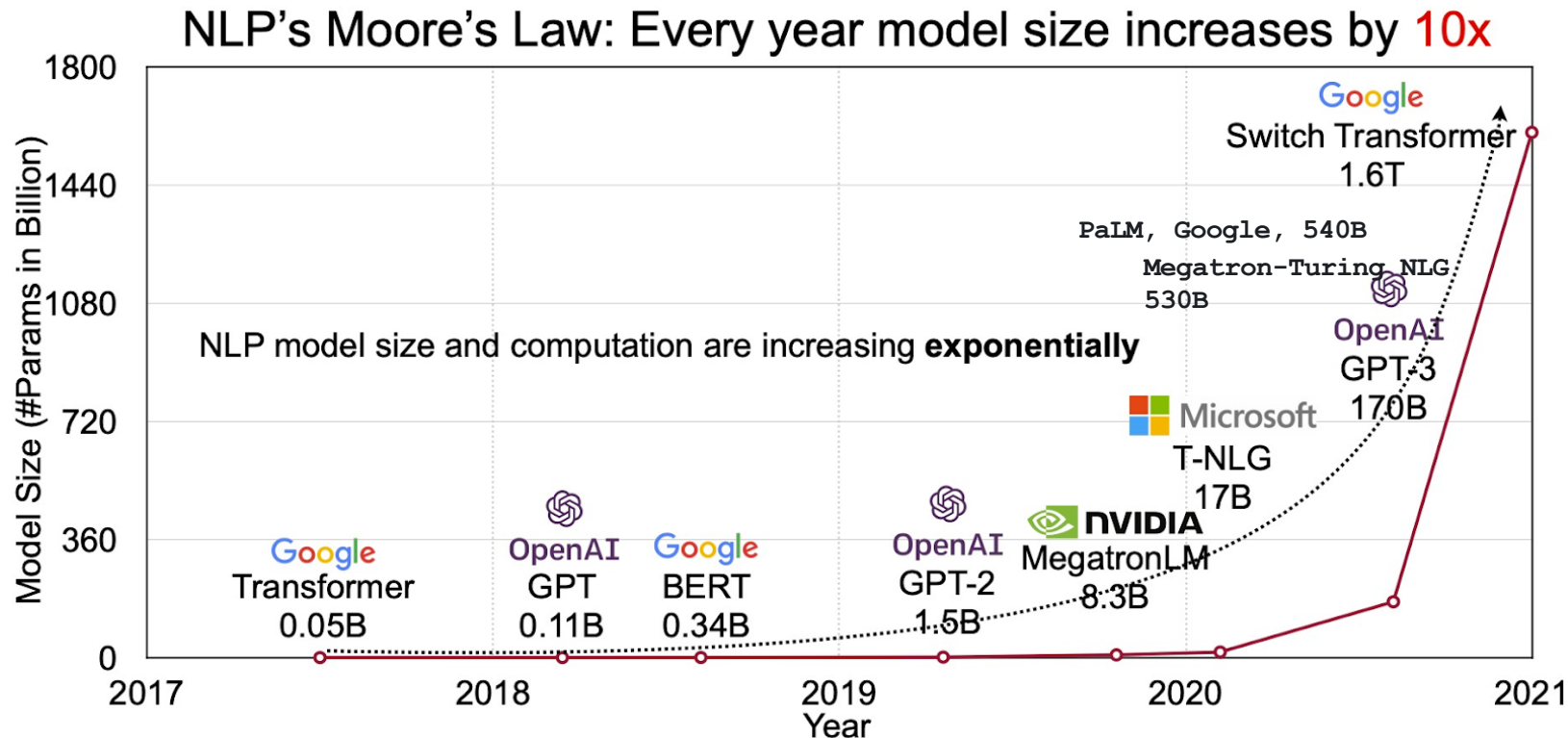


Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data

<http://www.economist.com/interactive/briefing/2022/06/11/huge-foundation-models-are-turbo-charging-ai-progress>

How large are large language models

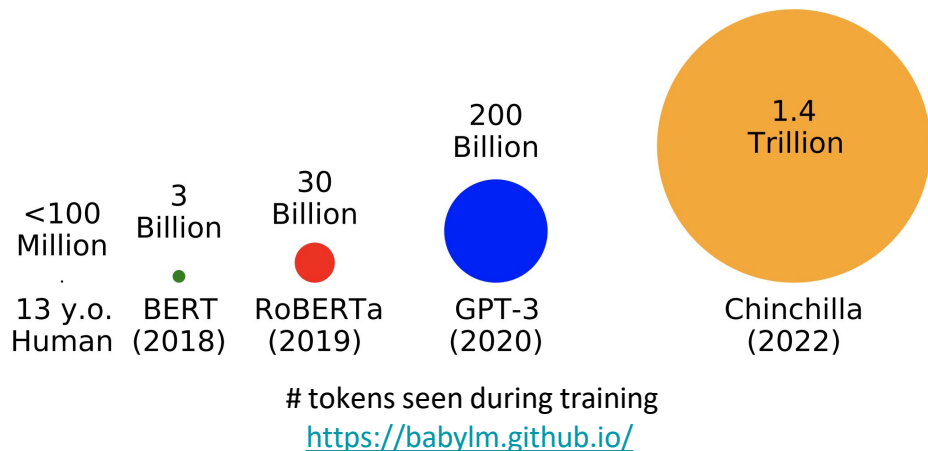
- Large model parameter size, large training data, large computing cost



Training data reaches Trillions

- To verify their findings, they trained an LLM (i.e. Chinchilla) using the same compute budget as Gopher, with much fewer model parameters (70B vs 280B) but much more training tokens (1.4T vs 300B).

Model	Size (# Parameters)	Training Tokens
LaMDA (thoppilan2022lamda)	137 Billion	168 Billion
GPT-3 (gpt3)	175 Billion	300 Billion
Jurassic (jurassic)	178 Billion	300 Billion
Gopher (rae2021gopher)	280 Billion	300 Billion
MT-NLG 530B (nlg530b)	530 Billion	270 Billion
Chinchilla	70 Billion	1.4 Trillion



Scaling law of LLMs (1)

- Scaling laws which are simple, predictive rules for model performance
- Empirical observation: scaling up models leads to reliable gains in perplexity
- Performance measures: cross entropy loss
- Resource measures: Model size, dataset size, computing power

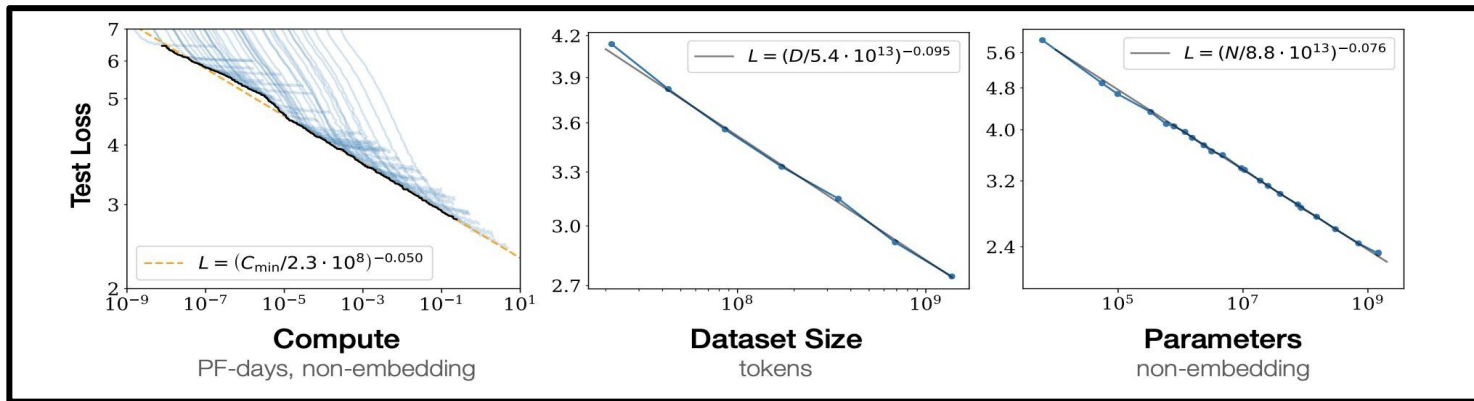
L – the cross entropy loss in nats. Typically it will be averaged over the tokens in a context, but in some cases we report the loss for specific tokens within the context.

N – the number of model parameters, *excluding all vocabulary and positional embeddings*

$C \approx 6NBS$ – an estimate of the total non-embedding training compute, where B is the batch size, and S is the number of training steps (ie parameter updates). We quote numerical values in PF-days, where one PF-day = $10^{15} \times 24 \times 3600 = 8.64 \times 10^{19}$ floating point operations.

D – the dataset size in tokens

Scaling law of LLMs (2)



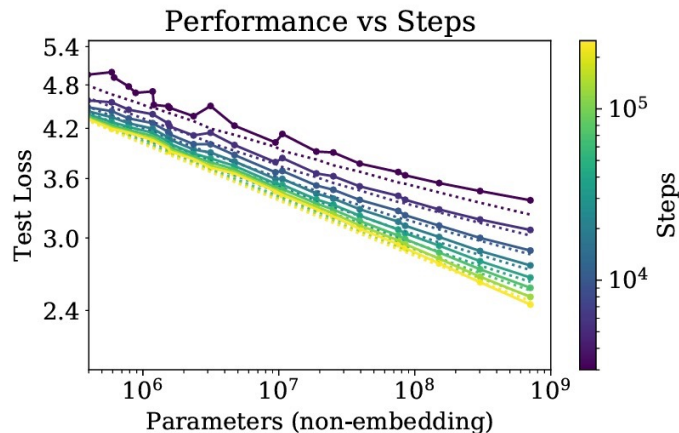
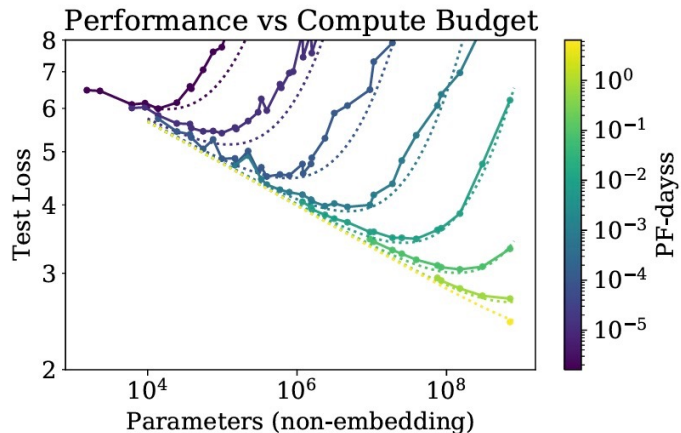
Performance depends strongly on scale, weakly on model shape: Model performance depends most strongly on scale, which consists of three factors: the number of model parameters N (excluding embeddings), the size of the dataset D , and the amount of compute C used for training. Within reasonable limits, performance depends very weakly on other architectural hyperparameters such as depth vs. width. (Section 3)

Smooth power laws: Performance has a power-law relationship with each of the three scale factors N, D, C when not bottlenecked by the other two, with trends spanning more than six orders of magnitude (see Figure 1). We observe no signs of deviation from these trends on the upper end, though performance must flatten out eventually before reaching zero loss. (Section 3)

Scaling law of LLMs (3)

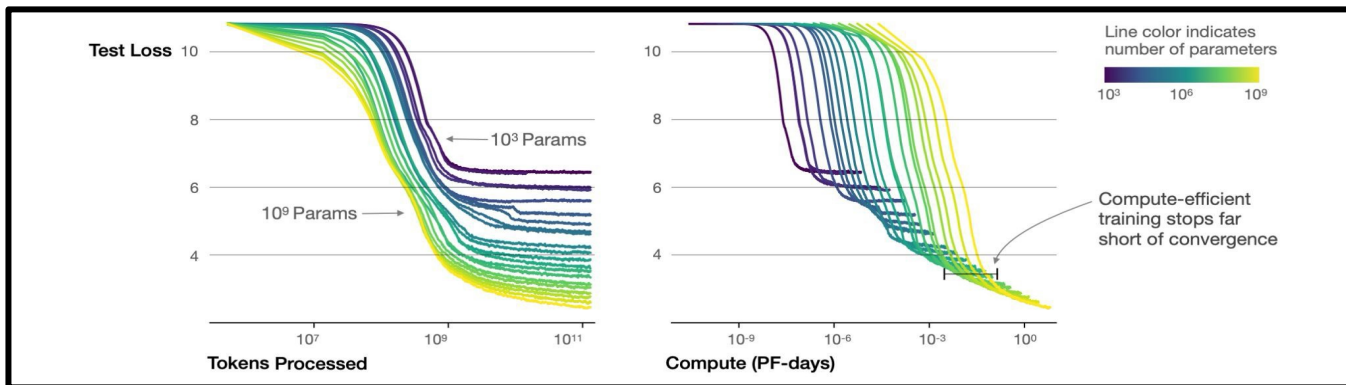
Universality of overfitting: Performance improves predictably as long as we scale up N and D in tandem, but enters a regime of diminishing returns if either N or D is held fixed while the other increases. The performance penalty depends predictably on the ratio $N^{0.74}/D$, meaning that every time we increase the model size 8x, we only need to increase the data by roughly 5x to avoid a penalty. (Section 4)

Universality of training: Training curves follow predictable power-laws whose parameters are roughly independent of the model size. By extrapolating the early part of a training curve, we can roughly predict the loss that would be achieved if we trained for much longer. (Section 5)



Scaling law of LLMs (4)

Sample efficiency: Large models are more sample-efficient than small models, reaching the same level of performance with fewer optimization steps (Figure 2) and using fewer data points (Figure 4).



Emergent abilities: zero-shot/few-shot learning

GPT-3 (175B parameters; [Brown et al., 2020](#))

- Another increase in size (1.5B -> **175B**)
- and data (40GB -> **over 750GB, Wikipedia, books, journals, Reddit links, Common Crawl, and other data**)
- Only training language models, amazingly achieve task-solving ability for other tasks (text generation, machine translation, reading comprehension)!

Zero shot may be too hard. Few shot learning (In-context learning) performs well!

1	gaot => goat	In-context learning ↓
2	sakne => snake	
3	brid => bird	
4	fsih => fish	
5	dcuk => duck	
6	cmihp => chimp	

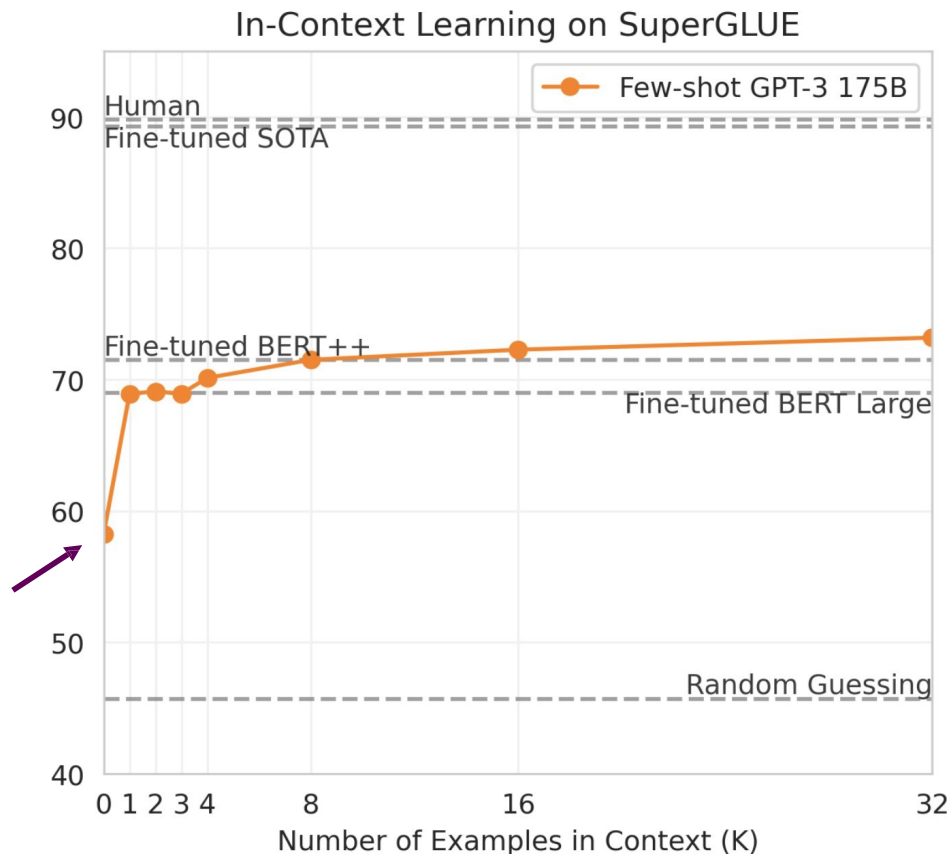
1	thanks => merci	In-context learning ↓
2	hello => bonjour	
3	mint => menthe	
4	wall => mur	
5	otter => loutre	
6	bread => pain	

Emergent abilities: zero-shot/few-shot learning (1)

Zero-shot

1 Translate English to French:

2 cheese =>

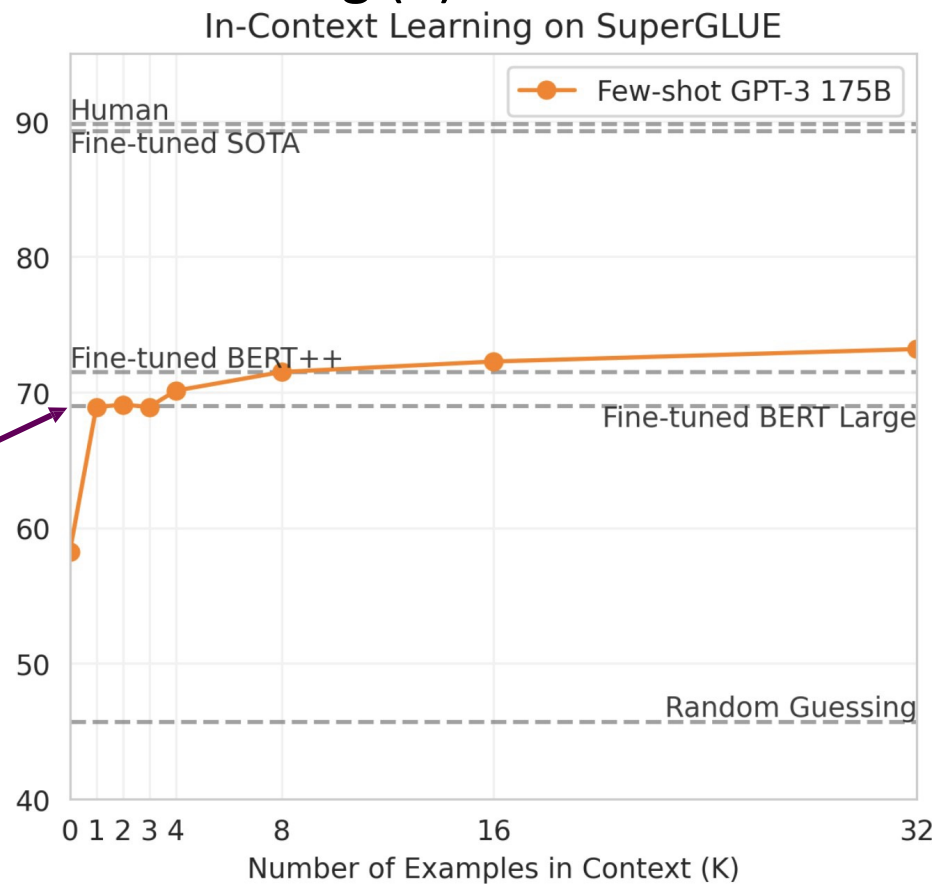


[Brown et al., 2020]

Emergent abilities: zero-shot/few-shot learning (2)

One-shot

1 Translate English to French:
2 sea otter => loutre de mer
3 cheese =>

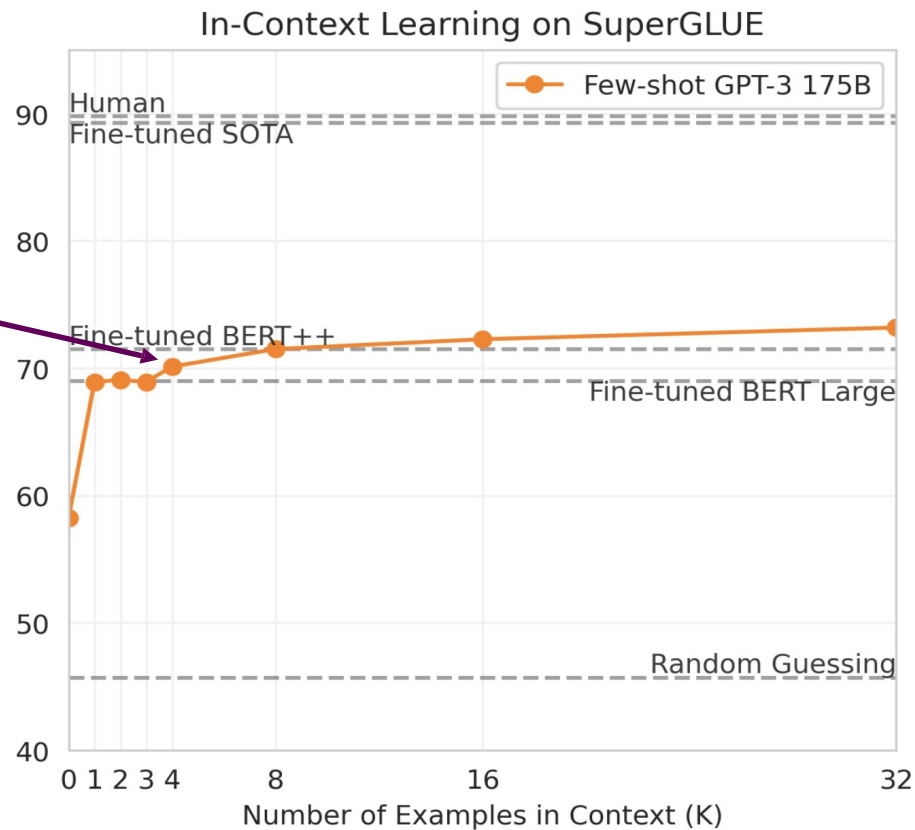


[Brown et al., 2020]

Emergent abilities: zero-shot/few-shot learning (3)

Few-shot

1 Translate English to French: ←
2 sea otter => loutre de mer ←
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ←



[Brown et al., 2020]

Emergent abilities: Chain-of-thought prompting (1)

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

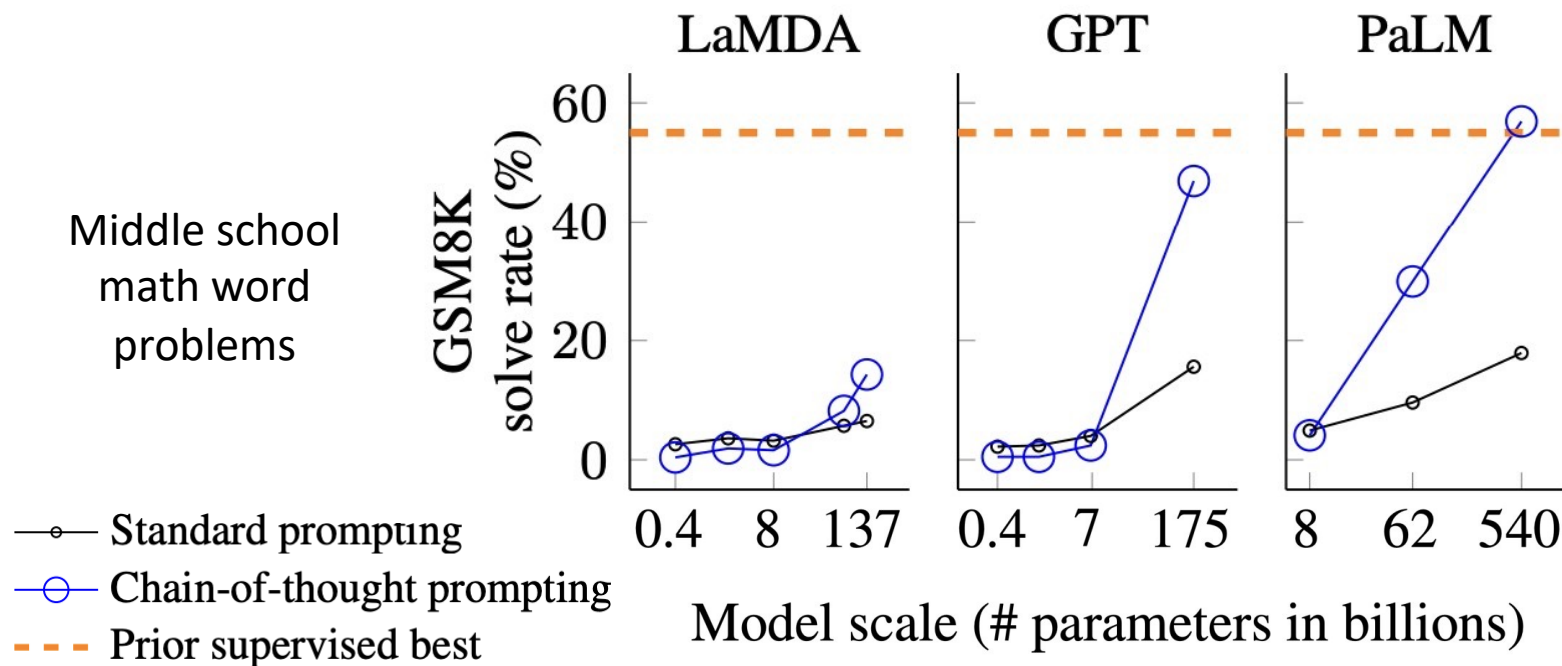
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Emergent abilities: Chain-of-thought prompting (2)



Language modeling $\neq$ assisting users

- Language models are not aligned with user intent [Ouyang et al., 2022].

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

Language modeling $\neq$ assisting users

- Language models are not aligned with user intent [Ouyang et al., 2022].
- **Finetuning to the rescue!**

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

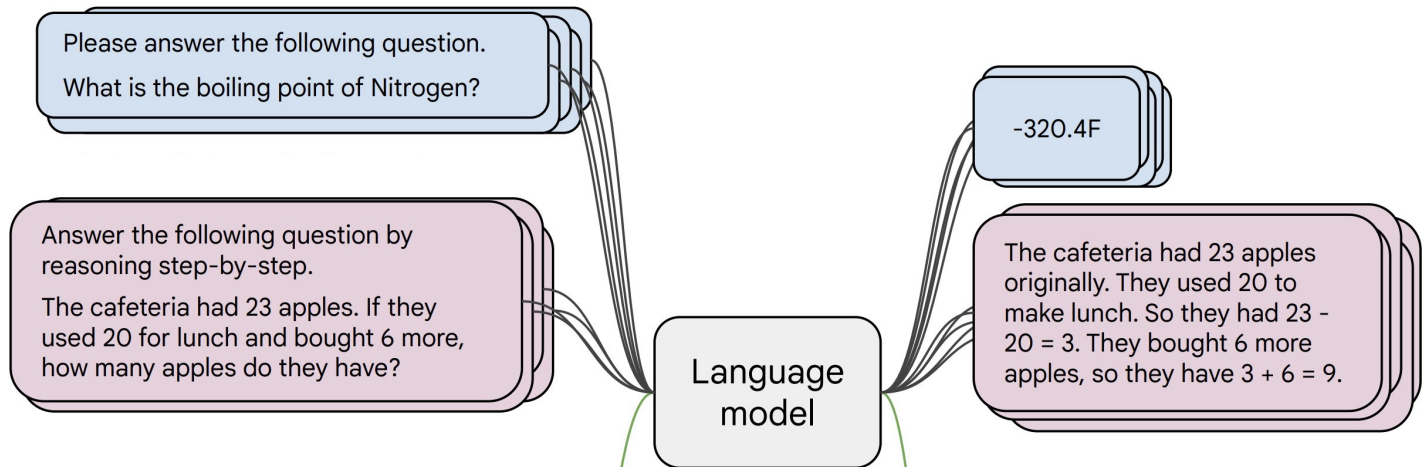
COMPLETION

Human

A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

Instruction finetuning

- **Collect examples** of (instruction, output) pairs across many tasks and finetune an LM



- **Evaluate on unseen tasks**



Without instruction finetuning

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:

- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

Before instruction finetuning

The reporter and the chef will discuss their favorite dishes.

The reporter and the chef will discuss the reporter's favorite dishes.

The reporter and the chef will discuss the chef's favorite dishes.

The reporter and the chef will discuss the reporter's and the chef's favorite dishes.

✗ (doesn't answer question)

Highly recommend trying FLAN-T5 out to get a sense of its capabilities:

<https://huggingface.co/google/flan-t5-xxl>

With instruction finetuning

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:

- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

After instruction finetuning

The reporter and the chef will discuss their favorite dishes does not indicate whose favorite dishes they will discuss. So, the answer is (C). ✓

Highly recommend trying FLAN-T5 out to get a sense of its capabilities:

<https://huggingface.co/google/flan-t5-xxl>

InstructGPT: Reinforcement Learning from Human Feedback

30k tasks

Step 1

Collect demonstration data, and train a supervised policy.

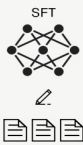
A prompt is sampled from our prompt dataset.

Explain the moon landing to a 6 year old

A labeler demonstrates the desired output behavior.

Some people went to the moon...

This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

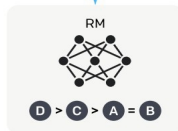
Explain the moon landing to a 6 year old

A Explain gravity...
B Explain war...
C Moon is natural satellite of...
D People went to the moon...

A labeler ranks the outputs from best to worst.

D > C > A = B

This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.

Write a story about frogs

The policy generates an output.



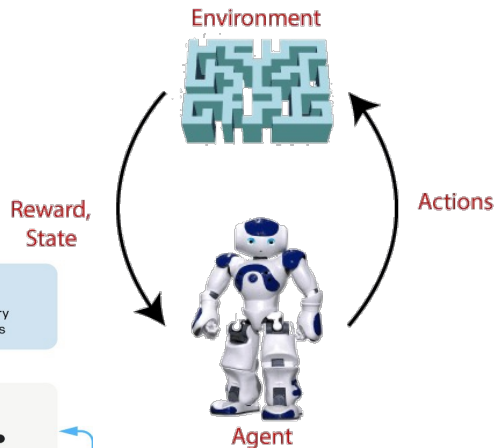
Once upon a time...

The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.

r_k



ChatGPT: Instruction Finetuning + RLHF for dialog agents

ChatGPT: Optimizing Language Models for Dialogue

Note: OpenAI (and similar companies) are keeping more details secret about ChatGPT training (including data, training parameters, model size)—perhaps to keep a competitive edge...

Methods

We trained this model using Reinforcement Learning from Human Feedback (RLHF), using the same methods as InstructGPT, but with slight differences in the data collection setup. We trained an initial model using supervised fine-tuning: human AI trainers provided conversations in which they played both sides—the user and an AI assistant. We gave the trainers access to model-written suggestions to help them compose their responses. We mixed this new dialogue dataset with the InstructGPT dataset, which we transformed into a dialogue format.

Summary

- Large language model definition
- Scaling law and Emergent ability
- Instruction tuning
- RLHF
- Challenges (not covered in the lecture)
 - Hallucination
 - Bias and fairness
 - Explainability
 - Privacy and data security